

$^{37}\text{Cl}(\text{d},\alpha\gamma)$ 1975VaYG,1972Va07

1975VaYG,1972Va07: A 4.25-MeV deuteron beam of 55 nA was provided from the Groningen 5 MV Van de Graaff generator.

The target was $100\text{ }\mu\text{g}/\text{cm}^2$ Co^{37}Cl enriched to 98% evaporated onto $10\text{-}\mu\text{g}/\text{cm}^2$ Formvar plus $10\text{-}\mu\text{g}/\text{cm}^2$ carbon. α particles were detected using a $60\text{-}\mu\text{m}$ annular silicon detector. γ rays were detected using a 120 cm^3 Ge(Li) detector at 90° . Measured $\sigma(E_\alpha)$, E_γ , I_γ , and $\alpha\gamma$ -coin. Deduced levels, γ -ray branching ratios. Also see 1971VaZG.

1968Te06: 3.1-4.6-MeV deuteron beams of 50 nA were provided from the Groningen 5-MV Van de Graaff generator. The target was a $100\text{-}\mu\text{g}/\text{cm}^2$ Co^{37}Cl both of natural ^{37}Cl abundance and enriched to 93% evaporated onto $10\text{-}\mu\text{g}/\text{cm}^2$ Formvar plus $10\text{-}\mu\text{g}/\text{cm}^2$ carbon. α particles were detected using an annular solid-state detector at $168\text{-}173^\circ$. γ rays were detected using a 3 in. by 3 in. NaI(Tl) scintillator at 55° . Measured $\sigma(E_\alpha)$, E_γ , and $\alpha\gamma$ -coin. Deduced levels.

1955Pa54: $^{37}\text{Cl}(\text{d},\alpha)^{35}\text{S}$ with 3.0, 5.6, 6.0, 7.0, and 7.5-MeV deuteron beams from the MIT-ONR electrostatic generator. Targets were 80 and $300\text{-}\mu\text{g}/\text{cm}^2$ barium chloride (75.4% ^{35}Cl , 24.6% ^{37}Cl) evaporated onto Formvar films on a gold layer. Charged reaction products emitted at 90° were magnetically analyzed by a broad-range spectrograph. Measured $\sigma(E_\alpha)$. Deduced levels.

 ^{35}S Levels

E(level) [†]	Comments
0	
1572.2 12	
1990.0 10	E(level): Other: 1992 10 from 1955Pa54.
2348.2 10	E(level): Other: 2348 10 from 1955Pa54.
2716.7 10	E(level): Other: 2714 10 from 1955Pa54.
2939.2 13	
3423 5	
3560.8 19	
3598.4 21	
3803.6 19	
3818.1 11	
3889.0 19	
4022.2 22	
4027.7 10	E(level): Other: 4025 10 from 1955Pa54.
4108 3	
4180 3	
4187 3	
4302 4	
4480.0 16	

[†] From 1975VaYG based on γ -ray energies, which however are not explicitly listed by the authors.

 $\gamma(^{35}\text{S})$

$E_i(\text{level})$	E_γ [†]	I_γ [‡]	E_f	Comments
1572.2	1572.2	100	0	
1990.0	1989.9	100	0	
2348.2	776.0	25 2	1572.2	
	2348.1	75 2	0	
2716.7	2716.6	100	0	
2939.2	2939.1	100	0	
3423	3423	100	0	
3560.8	1212.6	35 4	2348.2	
	1570.8	65 4	1990.0	
3598.4	3598.2	100	0	I_γ : >95 in 1975VaYG.
3803.6	2231.3	38 3	1572.2	
	3803.4	62 3	0	
3818.1	1828.1	100	1990.0	
3889.0	1540.8	40 4	2348.2	
	1898.9	45 5	1990.0	

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$^{37}\text{Cl}(\text{d},\alpha\gamma)$ 1975VaYG,1972Va07 (continued) $\gamma(^{35}\text{S})$ (continued)

<u>$E_i(\text{level})$</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>E_f</u>	<u>Comments</u>
3889.0	3888.8	15 3	0	
4022.2	2032.1	100	1990.0	
4027.7	1088.5	33 4	2939.2	
	1679.5	33 4	2348.2	
	2455.4	34 6	1572.2	
4108	4108	100	0	I_γ : >95 in 1975VaYG.
4180	2608	18 5	1572.2	
	4180	82 5	0	
4187	1839	100	2348.2	I_γ : >95 in 1975VaYG.
4302	1954	59 5	2348.2	
	4302	41 5	0	
4480.0	1763.3	36 4	2716.7	
	2489.9	15 4	1990.0	
	2907.7	39 4	1572.2	
	4479.7	10 2	0	

[†] Deduced by evaluators from level-energy difference in 1975VaYG.[‡] From 1975VaYG.

³⁷Cl(d,αγ) ¹⁹⁷5VaYG,1972Va07

Level Scheme

Intensities: % photon branching from each level

